

College Physics II

Lecture summaries

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Abstract

Disclaimer: These notes *summarize* the key concepts and formulae discussed during lecture. They should be supplemented by any standard algebra-based physics textbook on these topics (e.g., by Rex and Wolfson) or internet resource to fill in any gaps. Please send any comments, criticisms, suggestions to: joseph.d.romano@gmail.com.

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1 Electric charges, forces, and fields

1.1 Electric charge

- Properties of electric charge:
 - (i) Two types: *positive* and *negative* (+ and −), whose effects tend to cancel. Opposite charges attract, like charges repel.
 - (ii) Total electric charge is *conserved* (cannot be created or destroyed, only transferred from one object to another).
 - (iii) Electric charge is *quantized* (integer multiples of the charge of a proton $e = 1.602 \times 10^{-19}$ C; an electron has charge $-e$).
- Materials:
 - (i) *Conductors*: one or more electrons per atom are able to move freely through the material. Examples: aluminum, copper, gold, ... (metals).
 - (ii) *Insulators*: electrons are bound to atomic nuclei. Examples: rubber, wood, paper, ...
- Some demos:
 - 1) A plastic rod rubbed with fur is negatively charged; a glass rod rubbed with silk is positively charged. Show that opposite charges attract, like charges repel.
 - 2) Bits of uncharged paper will be attracted to both negative and positive rods due to charge separation within the atoms / molecules in the paper (an insulator).
 - 3) A charged rubber balloon sticks to the white board for the same reason as (2).

1.2 Electric force

- *Coulomb's law*: The magnitude of the electric force between two point charges q_1 and q_2 separated by a distance r is

$$F = \frac{k|q_1||q_2|}{r^2} \quad \text{where} \quad k = 8.99 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2} \quad (1)$$

- The force is attractive (directed towards one another) if the two charges have opposite signs; the force is repulsive (directed away from one another) if the two charges have the same signs.
- One sometimes writes

$$k = \frac{1}{4\pi\epsilon_0}, \quad \text{where} \quad \epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N} \cdot \text{m}^2} \quad (2)$$

where ϵ_0 is the so-called *permittivity* of free space (vacuum).

- The electric force on point charge Q due to a set of point charges $q_1, q_2, \dots, q_N$ is given by the vector sum of the forces

$$\vec{F}_{Q,\text{net}} = \vec{F}_{1Q} + \vec{F}_{2Q} + \dots + \vec{F}_{NQ} \quad (3)$$

where $\vec{F}_{1Q}$ is the electric force that point charge q_1 exerts on Q , etc. This is called the *superposition principle*.

1.3 Electric field

- The electric force between charges is a *non-contact* force. A set of charges creates a *field* to which another charge responds.

- The electric field at a point P in space is a vector $\vec{E}(P)$, whose direction is given by the electric force $\vec{F}_{q_0}(P)$ on a small, positive (so called “test”) charge q_0 placed at P , and whose magnitude is given by $F_{q_0}(P)/q_0$:

$$\vec{E}(P) = \vec{F}_{q_0}(P)/q_0 \quad (4)$$

Note that by dividing by q_0 , the electric field does not depend on properties of the test charge. It depends only on the charges producing the field.

- For a single point charge q :

$$E(P) = \frac{kq}{r^2} \quad (5)$$

where point P is a distance r from q . The direction of the electric field is radially away from the charge if $q > 0$ and towards the charge if $q < 0$.

- The electric field produced by a set of discrete point charges $q_1, q_2, \dots, q_N$ is given by the superposition principle:

$$\vec{E}(P) = \vec{E}_1(P) + \vec{E}_2(P) + \dots + \vec{E}_N(P) \quad (6)$$

where $\vec{E}_1(P)$ is the electric field due to q_1 , etc.

1.4 Electric field examples

- Electric dipole: Two charges, $+q$ and $-q$ separated by distance a along the x -axis (located at $x = -a/2$, $x = a/2$, respectively). Then the electric field at point $P = (0, y)$ on the y -axis is given by:

$$E_y(P) = 0, \quad E_x(P) = \frac{kqa}{[y^2 + (a/2)^2]^{3/2}} \quad (7)$$

When $y \gg a$, the electric field has magnitude $E(P) \simeq kqa/y^3$, which falls off as the *cube* of the distance to the dipole.

- The electric field produced by an infinite 2-dimensional sheet of charge with surface charge density σ has magnitude

$$E(P) = \frac{|\sigma|}{2\epsilon_0} \quad (8)$$

The field points perpendicular to the sheet (away from the sheet if $\sigma > 0$; toward the sheet if $\sigma < 0$). The magnitude of the electric field is independent of the distance from the sheet.

- In practice, one doesn't have 2-dimensional sheets that are infinite in extent. But the above results still hold for a finite sheet provided the point P is sufficiently close to the sheet.
- The electric field between two infinite, parallel sheets of charge with surface charge densities $+\sigma$ and $-\sigma$ has magnitude

$$E(P) = \frac{\sigma}{\epsilon_0} \quad (9)$$

The electric field points from the positively-charged sheet to the negatively-charged sheet.

- This last example will be important when we discuss *capacitors* later on.

1.5 Electric field lines

- Pictorial representation of the electric field $\vec{E}$. Direction of field lines point in the direction of $\vec{E}$; density of field lines (number per area perpendicular to $\vec{E}$) is proportional to the magnitude of $\vec{E}$.

- Rules for drawing electric field lines:
 - (i) lines begin on + charges, end on – charges (or may extend to infinity);
 - (ii) the number of lines leaving or terminating on a charge should be proportional to the magnitude of the charge;
 - (iii) electric field lines cannot cross.
- For a point charge, the density of lines at a distance r from the charge is given by $N/4\pi r^2$, which is proportional to the magnitude of the field $E(P) = kq/r^2$ as it should be.

1.6 Conductors in electrostatic fields

- Properties:
 - 1) $\vec{E} = \vec{0}$ everywhere inside a conductor, for both solid conductors and conductors with *empty* cavities.
 - 2) If an isolated conductor carries a charge, then that excess charge resides on the outer surface of the conductor.
 - 3) The electric field just outside the surface of a charged conductor is perpendicular to the surface with magnitude $E = \sigma/\epsilon_0$, where σ is the surface charge density.
 - 4) For irregularly shaped conductors, the surface charge density σ is greatest where the radius of curvature of the surface is smallest (i.e., where the surface is sharpest).
- Property (1) arises since charges in the conductor rearrange themselves in response to an applied electric field such that the *induced* field associated with these charges exactly cancels the applied field in the conductor, $\vec{E} + \vec{E}_{\text{ind}} = \vec{0}$.
- Property (1) implies that any external electric field is shielded from a cavity inside a conductor. That's why you're safe inside a car during a lightning storm.
- Fields within a cavity, produced e.g., by a charge, do make it outside of the conductor. So the outside of a conductor is not shielded from the fields inside a cavity of a conductor.

2 Electrostatic energy, potential, and capacitors

2.1 Review: Work and potential energy

- The *work* done by a force in moving a particle from point A to point B along some path is given by

$$W_{A \rightarrow B} = \sum_{A \rightarrow B} F_{\parallel} \Delta s = \sum_{A \rightarrow B} (F_x \Delta x + F_y \Delta y + F_z \Delta z) \quad (10)$$

where Δs is magnitude of the infinitesimal displacement vector $\Delta \vec{s}$ along the path and $F_{\parallel}$ is the component of the force $\vec{F}$ parallel to $\Delta \vec{s}$. The MKS unit of work is a Joule ($1 \text{ J} \equiv 1 \text{ N/m}$).

- For a general force, the work done depends on the path connecting A and B . If the force is *conservative*, then the work is path independent and depends only on the endpoints.
- Examples of conservative forces are gravity, the spring force, and the electrostatic force. An example of a non-conservative force is friction.
- For conservative forces, one can associate a *potential energy* U with the work done by the force in moving the particle from A to B :

$$\Delta U \equiv U_B - U_A = -W_{A \rightarrow B} = - \sum_{A \rightarrow B} (F_x \Delta x + F_y \Delta y + F_z \Delta z) \quad (11)$$

- Note that changing the potential energy U by an additive constant ($U' = U + \text{const}$) does not change the difference in potential energy, since $\Delta U = \Delta U'$. So the zero (or reference point) of the potential energy can be set in whatever way is most convenient for the problem at hand. (We will see that for point charges, the reference point is taken to be infinity.)
- From Equation (11) for ΔU , we see that the components of the conservative force $\vec{F}$ and the potential energy U are related by

$$F_x = -\Delta_x U, \quad F_y = -\Delta_y U, \quad F_z = -\Delta_z U \quad (12)$$

where $\Delta_x U$ denotes the change in the potential energy U as one moves a distance Δx in the x -direction, and similarly for y and z .

- Hence, we can calculate the force $\vec{F}$ by evaluating how the potential energy U changes in different directions in space.
- Note the minus sign in Equation (11). This means that the potential energy of the system decreases if the force does positive work. For example, as a mass m falls in a gravitational field $\vec{g}$, it loses potential energy, while gaining kinetic energy, $\Delta K = -\Delta U$.
- This last result implies that the total mechanical energy $E \equiv U + K$ of a system is unchanged (i.e., *conserved*) if only conservative forces are acting:

$$\Delta E = \Delta K + \Delta U = 0 \quad (\text{for conservative forces}) \quad (13)$$

2.2 Electrostatic potential energy

- Example: Consider three charges q_1, q_2, q_3 separated by distances r_{12}, r_{13}, r_{23} . The electrostatic potential energy associated with this system of charges can be found by calculating how much work an external agent has to do to “build” the configuration by bringing in charges one-by-one from infinity.
- You first bring q_1 to its position against no field; then q_2 to its position against the field created by q_1 ; and finally q_3 to its position against the field created by q_1 and q_2 :

$$U = \frac{kq_1q_2}{r_{12}} + \frac{kq_1q_3}{r_{13}} + \frac{kq_2q_3}{r_{23}} \quad (14)$$

- The above expression can be extended to an arbitrary number of charges.

2.3 Electrostatic potential

- The above results relating work and potential energy apply as well to the electrostatic force acting on a charge q

$$\Delta U = -W_{A \rightarrow B} = - \sum_{A \rightarrow B} (F_x \Delta x + F_y \Delta y + F_z \Delta z) = -q \sum_{A \rightarrow B} (E_x \Delta x + E_y \Delta y + E_z \Delta z) \quad (15)$$

where we used $\vec{F} = q\vec{E}$ to obtain the last equality.

- Dividing the above equation by q , we obtain

$$\Delta V \equiv \frac{\Delta U}{q} = - \sum_{A \rightarrow B} (E_x \Delta x + E_y \Delta y + E_z \Delta z) \quad (16)$$

where V is called the *electrostatic potential*.

- The electrostatic potential V is thus the electrostatic potential energy U per unit charge (similar to the electric field $\vec{E}$ being the electrostatic force $\vec{F}$ per unit charge). The MKS unit of electrostatic potential is the volt (V) defined as $1 \text{ V} = 1 \text{ J/C}$.

- If $\vec{E}$ is everywhere orthogonal to the displacement $\Delta\vec{s}$, then $\Delta V = 0$. This means that the electric field $\vec{E}$ is perpendicular to an *equipotential* surface, which has $\Delta V = 0$.
- Note that the units of $\vec{E}$ can be expressed either as 1 N/C (from $\vec{E} = \vec{F}/q$) or as 1 volt/m (from $E_x = -\Delta_x V$, the change in V in the x -direction). Thus, 1 N/C = 1 volt/m.
- Examples of electrostatic potentials:
 - (i) For a uniform electric field $\vec{E}$, we have $\Delta V = -Ed < 0$ if we move in the same direction as $\vec{E}$. Thus, for $q > 0$, the electrostatic potential energy *decreases* as you move in the direction of the field.
 - (ii) For a point charge q at the origin, $V(r) = kq/r$, where we have chosen zero potential at $r \rightarrow \infty$.
 - (iii) For a set of discrete point charges:

$$V(P) = \frac{kq_1}{r_1} + \frac{kq_2}{r_2} + \frac{kq_3}{r_3} + \cdots + \frac{kq_N}{r_N} \quad (17)$$

2.4 Electrostatic potential associated with conductors

- Since $\vec{E} = \vec{0}$ everywhere inside a conductor, $\Delta V = 0$ for any path lying within the conductor. Thus, a conductor is an *equipotential*.
- If a conductor has an empty cavity, then $\vec{E} = \vec{0}$ inside the cavity (and also with the conductor).
Proof: If that wasn't the case, then one find a path from one side of the cavity to the other having $\Delta V \neq 0$, contradicting that a conductor is an equipotential.
- The surface charge density on the surface of a conductor has its greatest magnitude where the radius of curvature is smallest.
Proof: Model your conductor as two conducting spheres with radii r_1, r_2 (with $r_1 > r_2$) and charge q_1, q_2 connected by a metal wire. Since this configuration is an equipotential, $kq_1/r_1 = kq_2/r_2$, which implies $q_2/q_1 = r_2/r_1$. It then follows that

$$\frac{\sigma_2}{\sigma_1} = \frac{q_2}{4\pi r_2^2} \bigg/ \frac{q_1}{4\pi r_1^2} = \frac{q_2}{q_1} \frac{r_1^2}{r_2^2} = \frac{r_2}{r_1} \frac{r_1^2}{r_2^2} = \frac{r_1}{r_2} > 1 \quad (18)$$

2.5 Capacitors

- A capacitor is a circuit element whose primary function is to *store charge*.
- Given two conductors (of any shape and size) with charges $\pm Q$, respectively, the capacitance C is defined by

$$C \equiv Q/\Delta V \quad (19)$$

where ΔV is the potential difference between the positively and negatively-charged conductors.

- The value of C is independent of Q and ΔV , depending only on the geometrical arrangement of the conductors.
- Example: A parallel-plate capacitor, having area A and separation d has capacitance $C = \epsilon_0 A/d$.
- One can connect capacitors in *parallel* or *series*, or combinations of the two. Such a combination of capacitors can be reduced to a single *equivalent* capacitance.
- For a parallel combination of C_1 and C_2 :

$$C_p = C_1 + C_2 \quad (20)$$

The proof of this relation uses the fact that the potential differences are equal, $\Delta V_1 = \Delta V_2 = \Delta V$, while the charges add $Q = Q_1 + Q_2$.

- For a series combination of C_1 and C_2 :

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} \quad (21)$$

The proof of this relation uses the fact that the potential differences add, $\Delta V = \Delta V_1 + \Delta V_2$, while the charges are equal, $Q = Q_1 = Q_2$.

- To charge up a capacitor, one needs to do work against the electric field associated with the charge already deposited on the plates.
- If q is the current charge on the capacitor, then the associated potential difference of the plates is q/C . To move additional charge dq against that potential requires work $dW = dq q/C$.
- The total work W is given by the area under the potential-difference-vs-charge triangle:

$$W = \frac{1}{2} \frac{Q^2}{C} \quad (22)$$

- This work is the electrostatic potential energy U stored in the capacitor, and it can be written in several equivalent ways:

$$U_E = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} C (\Delta V)^2 = \frac{1}{2} Q \Delta V \quad (23)$$

- A *dielectric* is an insulator placed between the conductors of a capacitor.
- Alignment of polarized molecules within the dielectric creates an induced electric field that *opposes* the applied electric field associated with the free charges $\pm Q$ on the two conductors.
- This induced electric field reduces the total electric field between the conductors, leading to a reduced value of the potential difference ΔV , and a corresponding increase in $C = Q/\Delta V$.
- The increase in capacitance is encoded in the (dimensionless) dielectric constant $\kappa \geq 1$, defined by $C = \kappa C_0$, where C_0 is the capacitance in the absence of a dielectric.
- For vacuum, $\kappa = 1$; for air, $\kappa = 1.00059$; for paper, $\kappa = 3.7$; for a conductor, $\kappa = \infty$.